



elevate - elevator inspection management for the municipality

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1. Background

The Câmara Municipal de Aveiro (CMA) is responsible for ensuring that the elevators installed in the municipality are periodically inspected. Today the process is largely manual: condominiums and private owners submit inspection requests through a form (around one hundred requests per cycle), municipal staff compile a list of the pending requests and send it to the contracted inspection company, technicians carry out the inspections, and the resulting reports are sent back and filed. The data entered in the request form is not reused in the document that the municipality sends to the inspection company, so it is re-typed by hand, and the only way to filter requests is by date.

There is also no register of the elevators themselves: the municipality does not know how many exist, where they are, or when each one was last inspected. Two sources can be used to build that register, the daily collection of requests and the urban planning licences that authorised each installation. Without it, the legal periodicity of inspections cannot be enforced; when an inspection is three months overdue the current practice is an informal warning. This project proposes elevate, a name that joins elevator and validate, since its purpose is to make sure every elevator in the municipality is known and every inspection is validated on time. elevate is an open-source platform that keeps a register of elevators, receives requests, tracks the full inspection lifecycle shared with the inspection company, and automatically issues the official notice (ofício) and payment prompt when an inspection falls due, including the handling of failed inspections and on-site complaints.

2. Goals

- Develop a functional prototype of an elevator inspection management platform adapted to the procedures of the Câmara Municipal de Aveiro.
- Build and maintain a register of the elevators in the municipality, populated from incoming requests and from urban planning licences.
- Digitise the full inspection lifecycle, from the request form to the inspection report, so that the data flows without re-typing between owners, the municipality, and the inspection company.
- Enforce the inspection periodicity through automatic deadlines, official notices, and payment prompts, replacing the current informal warnings.
- Publish the platform as open source so that other municipalities with the same obligation can adopt it.



3. Main Functionalities

- Elevator Register: identification, location, building and owner (condominium or individual), installation licence, inspection history, and current status, with import from urban planning licences and from the daily request intake.
- Request Intake: online form for condominiums and owners to submit inspection requests, with validation, automatic linking to the elevator record, and multi-criteria search and filtering (status, location, due date, owner, result).
- Inspection Scheduling & Sharing: automatic generation of the request list for the inspection company, a shared portal or export where the company confirms dates, assigns technicians, and uploads reports.
- Periodicity & Notices: due-date calculation per elevator, reminders, automatic generation of the official notice (ofício) when an inspection is due or overdue, and a prompt for the associated fee payment.
- Non-conformity Handling: workflow for failed inspections (re-inspection deadline, interdiction notice) and for on-site complaints reported at the elevator, linked to the elevator record.
- Dashboard & Indicators: elevators by status, overdue inspections, requests per period, approval rate, and average time from request to report, for municipal staff and management.

4. Work plan

Phase 1 – Requirements & Planning (October)

- Map the current inspection process through interviews with the CMA technical unit and the inspection company, and collect the applicable legal requirements on inspection periodicity.
- Define platform scope, MVP priorities, and user requirements for the three user types (owner, municipal staff, inspection company).
- Set up development workflow, tools, open-source licence, and responsibilities.

Phase 2 – System Design (Early November)

- Design system architecture (backend, frontend, database, integration with the inspection company).
- Define the elevator data model, the inspection state machine, and the rules for due dates and notices.
- Draft wireframes and UI mockups for the request form, elevator register, and dashboard.

Phase 3 – MVP Implementation (Mid November – December)

- Develop the core platform with the elevator register and the request intake form.
- Deliver MVP in December with request-to-list generation for initial evaluation.

Phase 4 – Extended Implementation (January – March)

- Expand with inspection scheduling and report sharing, periodicity control, automatic notices and payment prompts, and non-conformity handling.



- Import the existing licence and request data; improve integration, usability, and performance.

Phase 5 – Testing & Validation (April)

- Perform thorough unit, integration, and acceptance testing.
- Conduct user testing sessions with municipal staff and the inspection company on a real request cycle.
- Apply improvements based on feedback.

Phase 6 – Deployment & Documentation (May)

- Deploy prototype in a controlled environment at the municipality.
- Deliver user documentation and training materials.
- Present final system and recommendations for future iterations.